

Module Breakdown

User Management Module:

(Siva Sathwik B cb.sc.u4cse23110)
This module handles user authentication, registration, profile management, and role-based access (Student, Guardian, Admin).

File Path	Description
backend/routes/auth.js	Backend authentication routes (Login/Logout)
backend/routes/users.js	Backend user management routes
backend/models/User.js	User data model (Schema)
src/pages/Register.jsx	User registration page
src/pages/Profile.jsx	User profile management page
src/components/ProtectedRoute.jsx	Route guard for authenticated users
auth.test.js	Unit tests for authentication logic

Content & Course Management Module:

(Gayas Sheik cb.sc.u4cse23118)
This module is responsible for the creation, storage, and administrative oversight of educational content and courses.

File Path	Description
backend/routes/courses.js	Backend routes for course CRUD operations
backend/models/Course.js	Course data model (Schema)
src/pages/AdminDashboard.jsx	Administrative interface for content management
src/data/course_data.js	Static course content and metadata
backend/check_courses.js	Utility to validate course data integrity
backend/seed.js	Script to populate the database with initial content

Accessibility & Assistive Features Module:

(Yerneni Vishnu cb.sc.u4cse23154)

This module contains features specifically designed for learners with disabilities, including dyslexia-friendly settings and internationalization.

File Path	Description
src/context/AccessibilityContext.jsx	Global state for accessibility settings (fonts, colors)
src/pages/DyslexiaCenter.jsx	Dedicated hub for dyslexia-specific tools
src/styles/dyslexia-backdrop.css	Specialized styling for visual accessibility
src/locales/en.json	English localization strings
src/locales/es.json	Spanish localization strings
src/i18n.js	Internationalization configuration
src/utils/voice.js	Voice-to-text and assistive audio utilities
voice.test.js	Tests for voice-based assistive features

Learning Content Delivery Module:

(Naveen Paventha S cb.sc.u4cse23128)

This module focuses on the interactive delivery of lessons, quizzes, and specialized learning activities.

File Path	Description
src/pages/LessonPlayer.jsx	Main interface for delivering lesson content
src/pages/Quiz.jsx	Interactive quiz delivery system
src/pages/SpeechPractice.jsx	Specialized speech-based learning activity
src/components/PhonicsModule.jsx	Component for phonics-based learning
src/components/LetterTracing.jsx	Interactive letter tracing activity
src/components/RhymeLab.jsx	Interactive rhyming activity
src/components/MiniGames.jsx	Educational mini-games component
phonics.test.ts	Tests for phonics-based learning module

Analytics and Progress Tracking Module:

(V J Avinash cb.sc.u4cse23116)

This module tracks learner performance, XP, levels, and provides dashboard visualizations for students and guardians.

File Path	Description
src/pages/Dashboard.jsx	Student progress and analytics dashboard
src/pages/GuardianDashboard.jsx	Progress tracking interface for guardians
src/utils/gamification.js	Logic for XP, levels, and rewards
gamification.test.js	Tests for progress tracking and rewards logic